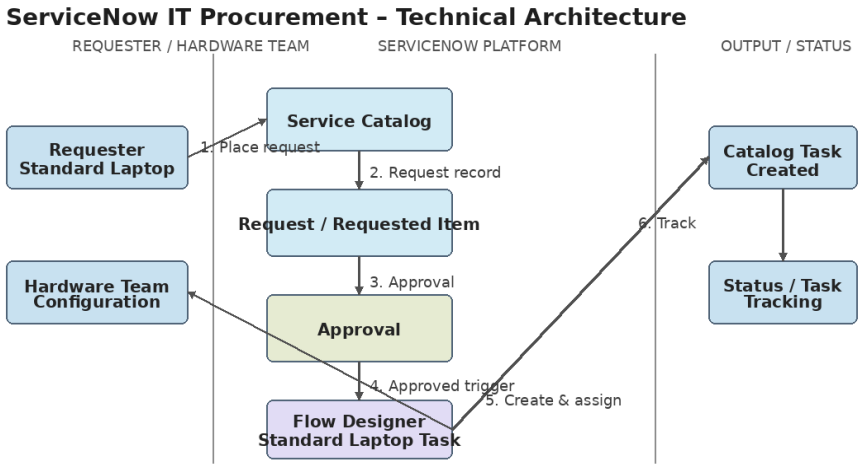
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 2025
Team ID	SWTID-2026-9795
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Architecture: ServiceNow Standard Laptop Procurement Automation
Project Platform: ServiceNow Flow Designer and Service Catalog



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	User interacts with the procurement request through ServiceNow Service Catalog.	ServiceNow Service Catalog
2.	Application Logic-1	Captures the approved service request and starts the automated task process.	ServiceNow Flow Designer
3.	Application Logic-2	Creates a Catalog Task for the requested Standard Laptop.	Flow Designer – Create Catalog Task action
4.	Application Logic-3	Updates task details and assigns the task to the Hardware group.	Flow Designer field values / Assignment group
5.	Database	Stores request, Requested Item and Catalog Task records used by the workflow.	ServiceNow platform records
6.	Cloud Database	Platform data used by the ServiceNow application.	ServiceNow platform data store
7.	File Storage	No separate file-storage component is required by the project workflow.	Not applicable
8.	External API-1	No external API is used in the implemented Standard Laptop workflow.	Not applicable
9.	External API-2	No external API is used in the implemented Standard Laptop workflow.	Not applicable
10.	Machine Learning Model	Machine learning is not part of this project implementation.	Not applicable
11.	Infrastructure (Server / Cloud)	The application and workflow run on the ServiceNow platform.	ServiceNow platform / cloud service

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No open-source framework is required for the implemented workflow.	Not applicable

S.No	Characteristics	Description	Technology
2.	Security Implementations	Approval and ServiceNow access controls are used; exact role/ACL configuration is not specified in the project.	ServiceNow access / approval controls
3.	Scalable Architecture	Flow Designer provides a reusable automation pattern that can be assigned to the Standard Laptop catalog item.	ServiceNow Flow Designer
4.	Availability	The procurement workflow is hosted within the ServiceNow platform and accessed through the Service Catalog.	ServiceNow platform
5.	Performance	Automation removes manual task creation and assignment after approval, reducing processing delay.	Flow Designer automation

References:

ServiceNow Service Catalog – project implementation reference
ServiceNow Flow Designer – project implementation reference
Create Catalog Task action – project implementation reference
Standard Laptop catalog item – project implementation reference
Hardware assignment group – project implementation reference